DESIGN AND SIMULATION OF A DIGITAL EVEN AND ODD PARITY GENERATOR USING XOR GATES

Course: Digital Logic / Digital Electronics
Project Type: Design and Simulation.

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INTRODUCTION TO PARITY

What is Parity?

- Parity is an error-detection technique used in digital communication.
- An additional bit is added to binary data.
- The parity bit makes the total number of 1s either:
  - Even → Even Parity
  - Odd → Odd Parity

Types of Parity

- **Even Parity:** Total number of 1s is made even.
- **Odd Parity:** Total number of 1s is made odd.

WHAT IS A PARITY GENERATOR?

Parity Generator

- A parity generator is a **combinational logic circuit** that generates a parity bit for a given set of binary inputs.
- The parity is calculated according to the number of 1s present in the input data.

EVEN PARITY

- The parity bit is selected so that the **total number of 1s become even**.

Example:

Data = 101

Number of 1s = 2

Even parity bit = 0

Therefore:

101 + 0 → 1010

Total number of 1s = **2**

ODD PARITY

- The parity bit is selected so that the **total number of 1s becomes odd**.

Example:

Data = 101

Number of 1s = 2

Odd parity bits = 1

Therefore:

101 + 1 → 1011

Total number of 1s = **3**

PROBLEM STATMENT

- Digital data can be affected by transmission errors.
- A simple method of detecting errors is parity checking.
- A parity generator adds an extra bit called the parity bit to the original data.
- The objective of this project is to design:
  - An Even Parity Generator
  - An Odd Parity Generator
- The complete circuit is designed and simulated using XOR gates in Logisim.

Objective:

To develop and verify a digital parity generator that produces the appropriate parity bit for a given binary input.

XOR GATE

Role of XOR Gate

- **XOR** stands for **Exclusive OR**

- An XOR gate produces an output of **1 when its inputs are different** and 0 when **they are the same**.

XOR TRUTH TABLE

A	B	A ⊕ B
0	0	0
0	1	1
1	0	1
1	1	0

WHY IS XOR USED?

- XOR is useful for parity generation because it produces **1 when there is an odd number of 1s** in its inputs.
- By connecting multiple XOR gates , parity can be generated for several input bits.

DESIGN METHODOLOGY

For a 3-bit input A, B, C :

Step 1: Apply the first two inputs to an XOR gate.

$$X1 = A \oplus B$$

Step 2: XOR the result with the third input.

$$X2 = X1 \oplus C$$

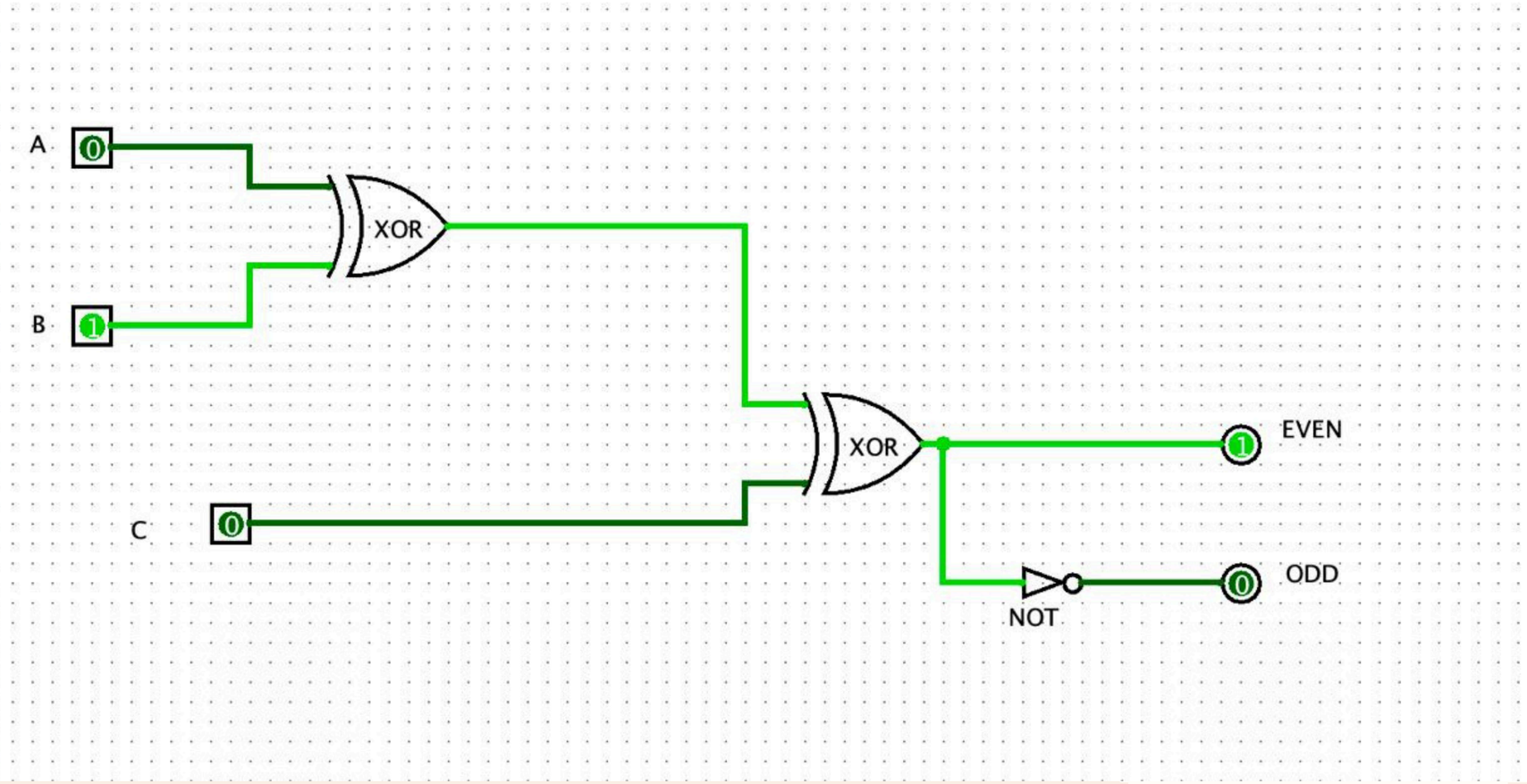
Therefore:

$$X = A \oplus B \oplus C$$

X = 1 → Input contains an odd number of 1s.

X = 0 → Input contains an even number of 1s.

CIRCUIT DESIGN



Components Used

- **3 Input Pins:** A, B and C
- **2 XOR Gates.**
- **1 NOT Gate.**
- **2 Output Pins.**

CIRCUIT OPERATION

First:

A ⊕ B = X

Then:

X ⊕ C = Even Parity

Therefore:

Even Parity = A ⊕ B ⊕ C.

The even parity output is connected to a NOT gate:

Odd Parity = ¬(A ⊕ B ⊕ C)

CIRCUIT FLOW

A, B

↓

XOR → X

↓

X, C

↓

XOR → Even Parity

↓

NOT → Odd Parity

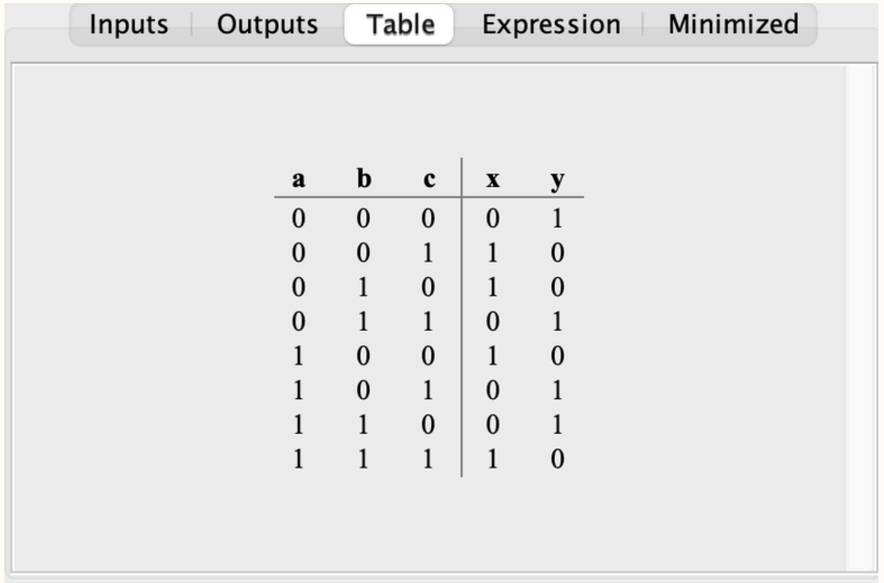
TRUTH TABLE

Truth Table of the Parity Generator

A	B	C	Even Parity	Odd Parity
0	0	0	**0**	**1**
0	0	1	**1**	**0**
0	1	0	**1**	**0**
0	1	1	**0**	**1**
1	0	0	**1**	**0**
1	0	1	**0**	**1**
1	1	0	**0**	**1**
1	1	1	**1**	**0**

OBSERVATION

- Even and odd parity outputs are always **complements of each other**.
- The circuit produces the expected output for **all 8 input combinations**



a	b	c	x	y
0	0	0	0	1
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	0

SIMULATION AND WORKING

Simulation in Logisim

The designed circuit was implemented and tested using **Logisim**.
Different combinations of inputs A, B, and C were given to verify the outputs.

Example Test Case

Input:
A = 0 ,B = 1,C = 0

The calculation is:
$0 \oplus 1 \oplus 0 = 1$.

Therefore:
Even Parity = 1
Since odd parity is the complement:
Odd Parity = 0

Result:
The simulated outputs matched the expected values in the truth table, confirming that the circuit works correctly.

CONCLUSION

- A digital even and odd parity generator was successfully designed.
- The circuit uses two XOR gates and one NOT gate.
- XOR gates are used to generate the even parity output.
- A NOT gate is used to obtain the complementary odd parity output.
- The circuit was successfully implemented and simulated in Logisim.
- All 8 possible input combinations were verified.
- The project demonstrates the use of digital logic for basic error detection.

GITHUB REPOSITORY

Project Repository

- Complete project files.
- Logisim circuit.
- Truth table.
- Simulation screenshots
- Project documentation

QR CODE

https://github.com/KLH-Parity-Generator/KLH-FED-2026-Session13-Batch-6

THANK YOU

ANY QUESTIONS?